Introduction to Software Engineering

COSC 336

ReSource (Phase III)

Requirements Document

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Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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This file contains the parts of the ReSource Phase 3 Requirements Document prepared by Mohamed Almarzooqi (100066203). Numbering, requirement IDs and cross-references are the same as in the combined group document.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS), version 1.0 for review, defines the required behavior of ReSource, an AI-powered campus resource exchange and asset life-cycle management prototype for Khalifa University. It specifies the functional requirements, quality requirements, permissions, business rules and use cases that will guide design, implementation and acceptance testing. Each requirement states an expected behavior and a way to check it; this document does not claim that the system has already been implemented or validated.

The requirements develop the course project description and brief [1], [2], the Phase 3 guidance [3] and the group's Phase 1 and Phase 2 work [4], [5]. Proposed university procedures require stakeholder confirmation. The interviews in Appendix E are explicitly AI simulations and are not evidence of actual university practice.

1.2 Document Conventions

"Shall" identifies a requirement. Functional requirements use FR followed by a feature and requirement number, such as FR-2.1. FR-7.N identifies notifications, FR-7.R identifies operational reports and FR-7.A identifies audit requirements. AI identifies the general AI requirements in Section 2.14. PER, SAF, SEC, QUA and BR identify performance, safety, security, quality and business-rule requirements. LEG and ORG identify legal and organizational requirements. UC identifies use cases and TBD identifies decisions awaiting confirmation (Appendix C).

Essential requirements are the ones the first working prototype must meet; together they cover the minimum prototype defined in the project description [1]. Desirable requirements are planned extensions, and Future requirements are outside the current minimum implementation. Each feature is given a High, Medium or Low priority, with a one-line reason; High features are part of that minimum prototype. Acceptance criteria are proposed tests, not completed test results. Every system requirement links to at least one of the use cases UC-1 to UC-13; the process requirements LEG-2, ORG-1, ORG-2 and ORG-4 and the policy rule BR-1 have no use case. Appendix D traces every requirement ID.

The document follows the SRS template in the project description [1]. As set out in the Phase 3 guidance [3], the template's Overall Description and External Interface Requirements sections are not required for Phase 3, so they are omitted and the remaining sections are numbered in order: Sections 2, 3 and 4 correspond to template Sections 4, 5 and 6. Role names follow the Phase 2 stakeholder roles [5] and are defined in Section 5.1.

1.3 Intended Audience and Reading Suggestions

Developers and testers should read the scope and roles first, then the functional requirements, business rules, state lists and linked use cases. They should use the acceptance criteria and Appendix D to plan implementation and tests. Reviewers of the specification can use Appendix D to check coverage, consistency and priorities against the project description [1].

Department Representatives, Asset Coordinators, Administrators, the Procurement Team, Maintenance Staff, Finance Officers and Sustainability Officers should review the workflows and permissions affecting their work. System Administrators and the IT Department should review

account management, security, auditability and unresolved technical policies. University Management and the proposed business owner, the Head of Facilities Management, should review scope, approval authority and Appendix C before authorizing any real pilot. The individual business owner and institutional approval remain unconfirmed.

The document is organized as follows. Section 2 specifies the thirteen system features (FR-1 to FR-13) and the general AI requirements that apply to every AI-assisted feature (Section 2.14). Section 3 sets measurable performance, safety, security and quality requirements and the business rules. Section 4 covers data, legal and organizational requirements. Section 5 defines the actors, the use case diagram and thirteen use case descriptions. Appendix A is a glossary, Appendix B lists the states of each record type, Appendix C lists open decisions, Appendix D traces every requirement and Appendix E records the AI-simulated stakeholder interviews.

1.4 Product Scope

ReSource will help authorized university departments publish surplus resources, find suitable assets and request reuse before considering a new purchase. It will record assets such as furniture, equipment, vehicles and books, including their condition, location, quantity, custody and life-cycle history. Controlled workflows cover requests, reservations, approvals, transfers, receipt, maintenance and approved end-of-life actions.

The prototype includes department-based access, consent at sign-up, notifications, operational reporting and sustainability indicators. AI supports classification, semantic matching, ranked suggestions with explanations and report drafting; sustainable-action recommendations and a natural-language assistant are planned extensions (Desirable). Authorized people review consequential decisions, and standard search, forms and reports remain available when AI cannot be used.

Expected benefits are better visibility of surplus assets, fewer avoidable purchases, longer useful asset life and clearer evidence of reuse. Savings and environmental indicators are estimates based on recorded data and stated assumptions; they are not measured campus outcomes until validated.

The academic prototype uses synthetic or reviewed sample data and existing AI tools. Live integration with university procurement, finance or single sign-on systems, autonomous purchasing or disposal, large-scale custom model training and production certification are outside the current scope [1], [5]. Physical transport, repairs and waste handling remain the responsibility of authorized people and service providers.

1.5 References

References [1] to [3] define the course requirements and the Phase 3 deliverable. References [4] and [5] are the group’s planning and feasibility baseline. References [6] to [10] identify UAE legislation used in the legal, security and data requirements; they are cited for university review and do not establish that the prototype is legally compliant. References [11] to [14] are university pages and technical standards used in specific requirements.

[1] Khalifa University, “Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System,” COSC 336 Introduction to Software Engineering project description, Khalifa Univ., Abu Dhabi, UAE, Fall 2026.

[2] Khalifa University, “COSC336 Circular Campus Project,” COSC 336 project brief slides, Khalifa Univ., Abu Dhabi, UAE, Fall 2026.

[3] Khalifa University, “Phase 3: Requirements Document,” COSC 336 lab slides, Khalifa Univ., Abu Dhabi, UAE, Fall 2026.

[4] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, “ReSource Phase I Report: Project Planning and Requirements,” COSC 336, Khalifa Univ., Abu Dhabi, UAE, Sep. 2026.

[5] R. Alzaabi, M. Alsuwaidi, A. AlAbdouli and M. Almarzooqi, “ReSource Phase II Feasibility Study,” COSC 336, Khalifa Univ., Abu Dhabi, UAE, Sep. 2026.

[6] United Arab Emirates, Federal Decree-Law No. 45 of 2021 Concerning the Protection of Personal Data, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1972

[7] United Arab Emirates, Federal Decree-Law No. 38 of 2021 on Copyright and Neighboring Rights, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1534

[8] United Arab Emirates, Federal Decree-Law No. 34 of 2021 on Countering Rumors and Cybercrimes, 2021. [Online]. Available: https://uaelegislation.gov.ae/en/legislations/1526

[9] United Arab Emirates Government, “Data protection laws,” u.ae, 2026. [Online]. Available: https://u.ae/en/about-the-uae/digital-uae/data/data-protection-laws

[10] Securiti, “Overview of UAE’s Federal Decree-Law No. (45) of 2021 on PDPL,” securiti.ai, 2026. [Online]. Available: https://securiti.ai/uae-personal-data-protection-law/

[11] Khalifa University, “Procurement & Contracts Department,” ku.ac.ae, 2026. [Online]. Available: https://www.ku.ac.ae/about/procurement-and-contracts-department

[12] Khalifa University, “Sustainability initiative,” ku.ac.ae, 2026. [Online]. Available: https://www.ku.ac.ae/sustainability/initiative

[13] OWASP Foundation, “OWASP Top 10:2025,” 2025. [Online]. Available: https://top10.owasp.org/2025/

[14] World Wide Web Consortium, “Web Content Accessibility Guidelines (WCAG) 2.2,” W3C Recommendation, Dec. 2024. [Online]. Available: https://www.w3.org/TR/WCAG22/

2. System Features

2.2 Resource Publication and Search (FR-2)

2.2.1 Description and Priority

This feature publishes surplus assets to the campus catalogue and lets Department Representatives find suitable resources through standard search and filters. Semantic matching is specified separately in FR-8; publishing and search remain usable without AI.

Priority: High. This feature develops Phase 2 requirement FR-2. Publication of available or surplus resources, search and filtering are part of the minimum prototype [1].

2.2.2 Stimulus/Response Sequences

1. The Asset Coordinator confirms publication of a complete asset record. The system checks the record, safety restrictions and permissions before making it visible.

2. The Department Representative opens the catalogue, enters a search and selects filters. The system validates the criteria and returns permitted matches with current availability.

3. The Department Representative opens an asset and reviews its details. If suitable, they continue to UC-3 to submit a request.

4. If no result matches, input is invalid or availability changes, the system explains the situation and allows the search to be refined.

2.2.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-2.1	The system shall publish an asset only when its Asset Coordinator confirms publication, the FR-1 fields are valid and no safety or life-cycle restriction blocks it (SAF-1, SAF-5, FR-5.9, FR-5.27). Drafts and withdrawn assets shall not appear in the shared catalogue.	Essential	UC-1, UC-2	A valid confirmed asset appears; a draft, withdrawn, unsafe, under-maintenance or pending end-of-life asset does not. Missing required fields prevent publication.
FR-2.2	The system shall let signed-in users search permitted published resources by asset ID, name, description or category using standard text search, without requiring AI.	Essential	UC-2	Known keywords and IDs return the expected permitted records while the AI service is disabled.
FR-2.3	The system shall combine filters for category, condition, owning department, location, required quantity, availability and life-cycle status. All selected filters shall apply together.	Essential	UC-2	A prepared data set returns only records meeting every selected filter. Clearing filters restores the permitted catalogue.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-2.4	The system shall show each result's asset ID, name, category, condition, department, location, available quantity and status, with a link to the permitted asset details.	Essential	UC-2	Every result shows these fields. Opening it shows the matching asset; restricted fields remain hidden.
FR-2.5	The system shall apply role and department permissions to results, counts, filters and direct record links. Financial values and value filters shall be available only to the roles authorized by SEC-5.	Essential	UC-2	A Department Representative can find shared assets but cannot view or infer restricted financial values or private records.
FR-2.6	The system shall validate search inputs, identify invalid ranges or quantities, and show a clear no-results message when nothing matches. Valid search criteria shall be retained after an error.	Essential	UC-2	A negative quantity or reversed range is rejected. A valid unmatched search shows zero results and allows refinement.
FR-2.7	The system shall show current availability when an asset is opened and allow a Department Representative to continue to UC-3. Browsing shall not create a request, reservation or approval.	Essential	UC-2, UC-3	Opening a result creates no request. If the last unit becomes unavailable, request submission is blocked under FR-3.7.
FR-2.8	The system shall support asset-age and estimated-value ranges where source data and permissions exist. Missing dates or values shall be shown as unknown, not treated as zero.	Desirable	UC-2	Known values filter correctly; unknown values are labeled and excluded from numeric ranges. Unauthorized users cannot apply value filters.

2.6 User Accounts, Access and Departments (FR-6)

2.6.1 Description and Priority

This feature controls who can use ReSource, which department they represent and which actions they may perform. The prototype uses local accounts; university single sign-on is outside the current scope. Consent is required at registration, with the notice wording and institutional policy awaiting confirmation (TBD-4).

Priority: High. This feature develops Phase 2 requirement FR-6. Account creation, authentication, role-based access control and department management are part of the minimum prototype [1].

2.6.2 Stimulus/Response Sequences

1. An Applicant reads the privacy notice and terms, enters the required details and explicitly consents. The system validates the application and records consent before retaining the registration (UC-13).

2. The System Administrator checks the application and recorded authorization, then activates the account with its permitted role and department (UC-8).

3. An active user signs in. The system authenticates the user and presents only authorized functions. Signing out ends the session.

4. The System Administrator changes an account, role or department. The system checks authority and dependencies, saves the change, audits it and notifies affected users.

5. A user requests a correction or withdraws consent. The system records the request; withdrawal stops account access and starts the approved data-handling process.

2.6.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-6.1	The system shall provide self-registration using name, university ID, university email, department and a password. It shall validate required fields and prevent duplicate university IDs or email addresses.	Essential	UC-13	A valid unique application is accepted. Missing fields, invalid credentials and duplicate identifiers are rejected without creating another account.
FR-6.2	Before saving a registration, the system shall display the current privacy notice and terms, explain the intended data uses and withdrawal route, and require an explicit, initially unticked consent choice. It shall record the account identifier, consent time and notice version.	Essential	UC-13	No account or registration data is retained when consent is absent. An accepted registration has a retrievable consent time and notice version.
FR-6.3	The system shall place new self-registrations in Pending approval. Only a System Administrator shall activate them and assign an approved role and department. Applicants shall not grant themselves privileges.	Essential	UC-8, UC-13	A new applicant cannot use asset functions or choose privileged access. Activation is recorded with the assigned role and department.
FR-6.4	The system shall let a System Administrator initiate an account invitation without retaining the invitee’s personal details before consent. The invitee shall provide their own details, password and consent before account activation.	Essential	UC-8, UC-13	An invitation alone creates no active personal account. Omitting consent prevents activation; consent is recorded on completion.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-6.5	The system shall authenticate active accounts, provide sign-out, and enforce the password, lockout and session rules in SEC-2 to SEC-4. Failed sign-in shall not grant access.	Essential	UC-1 to UC-13	Valid active credentials sign in. Incorrect credentials and pending, suspended or closed accounts cannot enter protected functions; sign-out invalidates the session.
FR-6.6	The system shall offer credential recovery only after verifying control of the registered account through an approved recovery process. It shall not reveal whether a submitted account identifier exists.	Essential	UC-13	Unverified recovery cannot reset credentials. Existing and unknown identifiers receive the same initial response. The recovery method is confirmed under TBD-3.
FR-6.7	The system shall check the user's current role, department and account status on every protected action and record access. Restricted actions shall remain blocked through direct links, reports and AI features.	Essential	UC-1 to UC-13	Each unauthorized role and department test is denied with no disclosure or state change; legitimate cross-department catalogue access still works.
FR-6.8	The system shall let only System Administrators assign, change or revoke roles and department membership, using recorded authorization and a reason. Users shall not change their own privileged access.	Essential	UC-8	An authorized change saves its approver and reason. Self-escalation and changes by other roles are refused and logged.
FR-6.9	The system shall let System Administrators create, rename and deactivate departments with unique IDs. Deactivation shall be blocked until active users, assets and open work are reassigned. Historical department references shall remain available.	Essential	UC-8	A duplicate ID is rejected. A department with unresolved dependencies cannot be deactivated; a clear department can, without losing history.
FR-6.10	The system shall let System Administrators suspend or close an account and revoke its active sessions. It shall preserve associated records and flag open work for reassignment to an authorized person. Suspending or closing the last active System Administrator account shall be blocked until another System Administrator account is active.	Essential	UC-8	The next protected action after suspension or closure is denied. Existing asset and request history remains; outstanding work is listed.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-6.11	The system shall let users review their account data, request corrections and withdraw consent. Withdrawal shall block further account use and notify the System Administrator to apply the confirmed retention and erasure policy.	Essential	UC-8	A correction or withdrawal request is recorded. Withdrawal ends active access; subsequent data handling follows the approved policy in SEC-8 and TBD-4.
FR-6.12	The system shall audit account, consent, role and department changes and notify affected users of access changes. Passwords and recovery secrets shall not appear in audit entries or notifications.	Essential	UC-8	Each tested change records actor, time and result; the affected user receives an appropriate notice without secrets.

2.7 Notifications, Reports and Audit Log (FR-7)

2.7.1 Description and Priority

This feature informs users of workflow events that need their attention, provides conventional, rule-based operational reports with fixed totals and tables, and keeps a complete, tamper-proof audit log of user and system actions. It refines Phase 2 requirement FR-7 and the project's requirements for notifications, operational reports and auditability. Notifications are in-application messages; email copies can be added once the delivery service is confirmed (TBD-6). Unlike the AI-generated reports in FR-12, the operational reports are deterministic: the same data and filters always give the same result, which makes them the reference source for every figure the system shows.

Priority: High. This feature develops Phase 2 requirement FR-7. Notifications and basic operational reports are part of the minimum prototype [1], and the audit log is required for security and legal compliance (3.3, 4.2).

2.7.2 Stimulus/Response Sequences

1. Workflow event recorded: a request, approval, transfer, maintenance, end-of-life or account event is saved → the system creates in-application notifications for the recipients in Table 2.7-1 and retries any failed delivery.
2. Report requested: an authorized user selects a report type and filters → the system generates the report from current data, limited to what the user may see.
3. Any recorded action**:** a user creates, changes, approves or rejects a record, signs in, is denied access, or uses an AI output → the system writes an audit entry automatically.
4. Audit search: a System Administrator or Administrator searches the audit log → the system lists the matching entries.

2.7.3 Functional Requirements

Notification requirements (FR-7.N)

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-7.N1	After a workflow event is successfully recorded, the system shall create an in-application notification for the responsible users listed in Table 2.7-1.	Essential	UC-3 to UC-9, UC-12, UC-13	Each listed test event creates the expected recipient notifications; a failed transaction creates no success notification.
FR-7.N2	Each notification shall identify the event, linked record, current status, time and required next action. Links and content shall respect the recipient's current permissions and exclude passwords and restricted personal or financial data.	Essential	UC-3 to UC-9, UC-12, UC-13	Every notice contains the required fields. A user whose access was revoked cannot open the linked protected record.
FR-7.N3	The system shall show required event notifications within five minutes of a committed event under the PER-1 test workload. A notification failure shall not reverse a valid, audited business action.	Essential	UC-3 to UC-9, UC-12, UC-13	All events in the notification test appear within five minutes. A forced sending failure leaves the saved decision and audit entry intact.
FR-7.N4	The system shall let signed-in users view their own notification history and mark notifications as read or unread. Reading a message shall not approve, accept or complete its linked action.	Essential	UC-3 to UC-9, UC-12, UC-13	A user sees only their own notices; changing read status makes no change to any business record.
FR-7.N5	The system shall retain failed notification deliveries for retry and make failures visible to a System Administrator. Retrying shall not create duplicate in-application messages for the same event and recipient.	Essential	UC-3 to UC-9, UC-12, UC-13	A simulated failure is visible; recovery produces exactly one in-application message per event and recipient.
FR-7.N6	When the desirable expiry and reminder rules are enabled, the system shall notify affected users when a request or reservation expires under FR-3.9 or a receipt remains unconfirmed for five working days (FR-4.13). Email copies may be enabled after the delivery service is confirmed.	Desirable	UC-3, UC-4, UC-5	Each enabled deadline generates one notice to the specified recipients. Calendar rules and email configuration match TBD-5 and TBD-6.

Table 2.7-1: Notification routing

Recorded event	Recipients

Recipients are de-duplicated where one person fills several permitted roles. A notification contains only information the recipient may see. Deadline rules use the confirmed working-day calendar (TBD-5). Inactive-account notices use the limited verified account-status route described in FR-6; email delivery remains subject to TBD-6.

3. Other Nonfunctional Requirements

3.5 Business Rules

These are proposed prototype rules derived from the project description and the group's workflows [1], [4], [5]. The university must confirm delegation, thresholds and local policy before a real pilot (TBD-1, TBD-2). The table states the responsible human decisions, the automated checks that support them and how each rule is checked. BR-3 defines the approval stages for resource requests and BR-8 the approval stages for end-of-life actions.

ID	Rule	How it is checked	Applies to	Priority
BR-1	The proposed owner of the reuse-first policy is the Head of Facilities Management, with Procurement, Finance and University Management confirming their delegated authority. This is a project proposal, not a verified university policy. The System Administrator configures approved rules but does not decide them.	Inspection: each configured rule records its approving authority (TBD-1).	Policy configuration; TBD-1, TBD-2	Essential
BR-2	Only the owning department's Asset Coordinator may register, publish or release its assets. A Department Representative may identify surplus and submit or track departmental requests. Publication requires complete data and a safe, eligible asset; it grants no transfer approval.	Test: registration, publication and release by any other role are refused (FR-1.1, FR-2.1, FR-4.1).	UC-1 to UC-4; FR-1, FR-2, SAF-5	Essential
BR-3	For an asset-specific request, the owning Asset Coordinator first releases or declines it. An Administrator then approves or rejects the released request. Decline and rejection require a reason. If FR-4.3 is enabled, a Procurement Team member must also approve values above the configured limit or with unknown value. All required approvals must exist before Approved status.	Test: a request cannot reach Approved while any required decision is missing.	UC-4; FR-4.1 to FR-4.4; TBD-2	Essential
BR-4	A person shall not approve their own request or both release and approve it. The same person cannot satisfy two required approval stages. Multiple assigned roles do not bypass these checks. System Administrator access provides no asset-transfer or disposal approval authority.	Test: 100% of self-approval and combined-stage attempts are refused and logged.	UC-4, UC-7, UC-8; FR-4.4; FR-6.8; SAF-3	Essential

ID	Rule	How it is checked	Applies to	Priority
BR-5	Submitting a request does not reserve units or change ownership, custody or location. The requested quantity is reserved only when the final required approval is recorded, after availability and permissions are checked again. Cancellation, rejection or expiry releases only the affected reservation, and no process may allocate more units than are available.	Test: submission leaves the available quantity unchanged; approval reduces it; simultaneous approvals never over-allocate.	UC-3 to UC-5; FR-3.7 to FR-3.9; FR-4.5	Essential
BR-6	An Asset Coordinator may arrange and record handover only for an approved request. A Department Representative or the Asset Coordinator of the receiving department confirms receipt and arrival condition. Ownership, custody and location change only after receipt. An Administrator resolves disputes; the resolution and any return are recorded.	Test: receipt by any other user is refused, and custody is unchanged before receipt.	UC-5; FR-4.7 to FR-4.16	Essential
BR-7	Maintenance Staff assess safety and recommend repair or return to service. Only Maintenance Staff or an Administrator may record inspection and repair results (FR-5.5, FR-5.6, FR-5.15); an Administrator cannot override a required safety inspection (SAF-5, SAF-9). Repairs or spending that need authorization must have a recorded human decision under the confirmed delegation. System-generated alerts, such as the maintenance-cost threshold (FR-5.11), do not authorize repair expenditure.	Test: inspection and repair results entered by other roles are refused; no alert changes a repair or spending decision.	UC-6; FR-5.5, FR-5.6, FR-5.11, FR-5.15; SAF-2, SAF-5, SAF-9; TBD-2, TBD-7	Essential

ID	Rule	How it is checked	Applies to	Priority
BR-8	End-of-life actions pass through two approval stages. Stage 1, safety review: a Sustainability Officer other than the proposer confirms or corrects the hazard classification and, for hazardous or electronic-waste items, assigns a handling party (FR-5.23 to FR-5.25). Stage 2, approval: an Administrator who is not the proposer approves or rejects the proposal with a reason (FR-5.26, FR-5.28). An Asset Coordinator or Sustainability Officer may propose the action with a justification; disposal also needs a reason why reuse, repair, donation and recycling are unsuitable (SAF-4). Completion needs the completion record and, for hazardous or electronic-waste items, the handling party's certificate (FR-5.29 to FR-5.32).	Test: an end-of-life action cannot reach its final state without both stages and a complete completion record.	UC-7; FR-5.20 to FR-5.34; SAF-1 to SAF-6; LEG-7; TBD-2, TBD-7	Essential
BR-9	AI may suggest classifications, matches, actions and text. An authorized person must confirm saved values and consequential actions. AI cannot grant access, allocate authority, approve spending, approve requests, change custody, certify safety or complete end-of-life actions. People may reject or correct suggestions; consequential overrides require a recorded reason.	Test: 0 record changes are caused directly by AI outputs (SAF-2).	UC-7, UC-9 to UC-12; AI-1 to AI-10	Essential
BR-10	The Procurement Team and the department's budget holder decide whether to buy new or reuse under university procedures [11]. ReSource supplies reuse evidence and records approved asset workflows; it does not place purchase orders or authorize purchases. Finance Officers review financial inputs; Sustainability Officers review environmental assumptions.	Review: no system function approves or blocks a purchase (LEG-6).	UC-9, UC-12; LEG-6	Essential
BR-11	An AI report must be approved before it is shared (FR-12.9). A report that contains financial figures may be approved only by a Finance Officer or an Administrator; a report without financial figures may also be approved by a Sustainability Officer. Distribution is limited to the information each recipient may see (SEC-5).	Test: a Sustainability Officer cannot approve a report with financial figures, and recipients without financial access see none.	UC-12; FR-12.9; SEC-5	Essential

ID	Rule	How it is checked	Applies to	Priority
BR-12	All approvals, refusals, overrides and access changes require an attributable record. Failed audit recording prevents a change from committing; a later notification failure does not undo a successfully committed decision. Completed history is corrected through linked records, not overwritten.	Test: with audit recording forced to fail, no change is saved (FR-7.A6).	UC-1 to UC-13; FR-7.A1 to FR-7.A6; SEC-12	Essential

5. Use Cases

5.1 Actors and Roles

The first ten entries are the roles that use the system directly. The first nine are the Phase 2 stakeholder roles [5], written in the singular, with their Phase 2 category in the Type column; the tenth, Applicant, is a person who has applied but has no active account yet. External participants support recorded activities but have no account in the prototype. AI matching, classification, recommendation, assistant and reporting components are part of ReSource and are not actors. The Head of Facilities Management (proposed business owner), the IT Department (proposed technical owner), developers and lab instructors are project stakeholders; they receive system permissions only if separately assigned an approved operational role. Appendix A maps the role names used in the project description to these roles. Table 5.1-1 summarizes who may perform each controlled action; every permission is enforced by the requirements named in the table.

Actor	Type	Main responsibilities and permissions
Department Representative	User	Identify departmental needs and surplus, and coordinate surplus listings with their Asset Coordinator; search shared resources; submit and track own-department requests; confirm receipt for their department; report defects; use the assistant. No approval of own requests. UC-2, UC-3, UC-5, UC-6, UC-11.
Asset Coordinator	User	Register and publish assets; review AI classification; release or decline requests; arrange transfers; confirm receipt for their department; report defects; propose end-of-life actions. UC-1, UC-4, UC-5, UC-6, UC-7, UC-10.
Administrator	Business user	Approve released requests and end-of-life proposals; resolve transfer disputes; assign maintenance work; handle human-review requests; maintain the approved vendor list; review permitted reports and audit records. No self-approval or technical role administration. UC-4, UC-5, UC-6, UC-7, UC-12.
Procurement Team	User	Review and override matches for unmatched needs; see authorized costs; give the second approval required by FR-4.3; request AI reports. Purchasing remains outside ReSource. UC-4, UC-9, UC-12.
Maintenance Staff	User	Record defects, inspection findings, repairs and safety assessments for assigned work. Repair-cost entry does not grant general financial-report access. UC-6.
Finance Officer	User	Maintain authorized replacement-price and cost assumptions; generate and review savings reports. No asset-transfer or disposal approval solely from this role. UC-12.
Sustainability Officer	User	Maintain indicator defaults; review sustainable-action advice; perform safety reviews and confirm hazard classification; propose end-of-life actions; review and approve AI reports without financial figures. UC-6, UC-7, UC-12.

Actor	Type	Main responsibilities and permissions
System Administrator	Technical user	Manage accounts, roles, departments, approved configuration and delivery failures; inspect permitted audit records. No business approval or self-escalation. UC-8, UC-13.
University Management	Customer (sponsor)	View authorized approved management reports. Budget and adoption authorization take place through university procedures. UC-12.
Applicant	Prospective user	Apply for an account, read the privacy notice and give or withhold consent. No access to asset functions until a System Administrator activates the account. UC-13.
Facilities Management	External operational participant	Provide manual review and handling evidence for UC-7 through authorized campus staff. No vendor portal or extra account privileges are assumed.
Approved Recycling Vendor	External operational participant	Supply handling and recycling or disposal certificates for UC-7, entered by an authorized campus user. No direct prototype login.
Donation Recipient	External operational participant	Provide recipient and handover evidence for UC-7 through authorized campus staff. No direct prototype login.

Table 5.1-1: Permissions by controlled action

Controlled action	Who may perform it	Conditions (requirement)
Register, edit, publish or withdraw an asset	Asset Coordinator of the owning department	Hazardous assets need a safety review before publication (FR-1.1, FR-1.15, FR-2.1).
Search the catalogue	All signed-in users	Only published, unrestricted assets; financial fields only as allowed by SEC-5 (FR-2.2, FR-2.5).
Submit, edit or cancel a request	Department Representative (own department)	Edit only while Submitted; cancel before handover (FR-3.5, FR-3.11).
Review AI matches for an unmatched need	Procurement Team member	The chosen asset still goes through release and approval (FR-8.7).
Release or decline a request	Asset Coordinator of the owning department	Not for a request the same person submitted (FR-4.1, FR-4.4).
Approve or reject a released request	Administrator; a Procurement Team member for the second approval	A different person for each stage; availability is rechecked before approval (FR-4.2 to FR-4.5).
Arrange a transfer and record handover	Asset Coordinator of the owning department	Approved requests only; safety conditions checked at handover (FR-4.7, FR-4.16).

Controlled action	Who may perform it	Conditions (requirement)
Confirm receipt	Department Representative or Asset Coordinator of the receiving department	Condition on arrival is required (FR-4.10).
Decide a transfer dispute	Administrator	Custody changes only after the decision (FR-4.12).
Report a defect or request an inspection	Asset Coordinator, Department Representative, Maintenance Staff	Inspection requests: Asset Coordinator or Maintenance Staff (FR-5.1, FR-5.2).
Record inspection and repair results	Assigned Maintenance Staff member; Administrator	All other roles are refused (FR-5.15).
Propose an end-of-life action	Asset Coordinator, Sustainability Officer	Justification required; disposal needs a reason for each alternative (FR-5.20, SAF-4).
Record the end-of-life safety review	Sustainability Officer who is not the proposer	Assigns a handling party for hazardous items (FR-5.23, FR-5.24, SAF-3).
Approve an end-of-life action	Administrator who is not the proposer	Second approval stage (BR-8).
Maintain the approved vendor list	Administrator	Only unexpired vendors can be assigned (FR-5.25).
Maintain category default values	Finance Officer (prices and costs); Sustainability Officer (weights, years of use, emission factors)	Every value needs a source and an effective date (FR-13.4).
Generate standard reports	Asset Coordinator, Department Representative, Procurement Team, Finance Officer, Sustainability Officer, Administrator, University Management	Own department unless the role covers all departments (FR-7.R1, FR-7.R3).
Request an AI report	Finance Officer, Sustainability Officer, Procurement Team member, Administrator	Scope limited to the user’s permitted data (FR-12.1, FR-12.3).
Approve an AI report for sharing	Finance Officer or Administrator; Sustainability Officer only for reports without financial figures	Drafts are never shared (FR-12.9, BR-11).
View financial information	Finance Officer, Procurement Team, Administrator, University Management; Asset Coordinator (own assets); Maintenance Staff (own work); Sustainability Officer (aggregates only)	SEC-5.

Controlled action	Who may perform it	Conditions (requirement)
Manage accounts, roles and departments	System Administrator	No business approvals and no self-escalation (FR-6.8, BR-4).
View and search the audit log	System Administrator, Administrator	Entries cannot be edited (FR-7.A2, FR-7.A3).
Apply for an account	Applicant	Consent required; activation by a System Administrator (FR-6.1 to FR-6.3).

5.2 Use Case Diagram

The combined diagram shows UC-1 to UC-13 and the ten roles that use the system. Lines link each use case to its initiating actor and to the main approving, reviewing or receiving roles; other supporting participation is described in the use case descriptions and Section 5.1. Lines do not show execution order or permission inheritance. UC-10 extends UC-1 when AI suggestions are requested. External handling participants are listed below the diagram because they supply evidence through campus users.

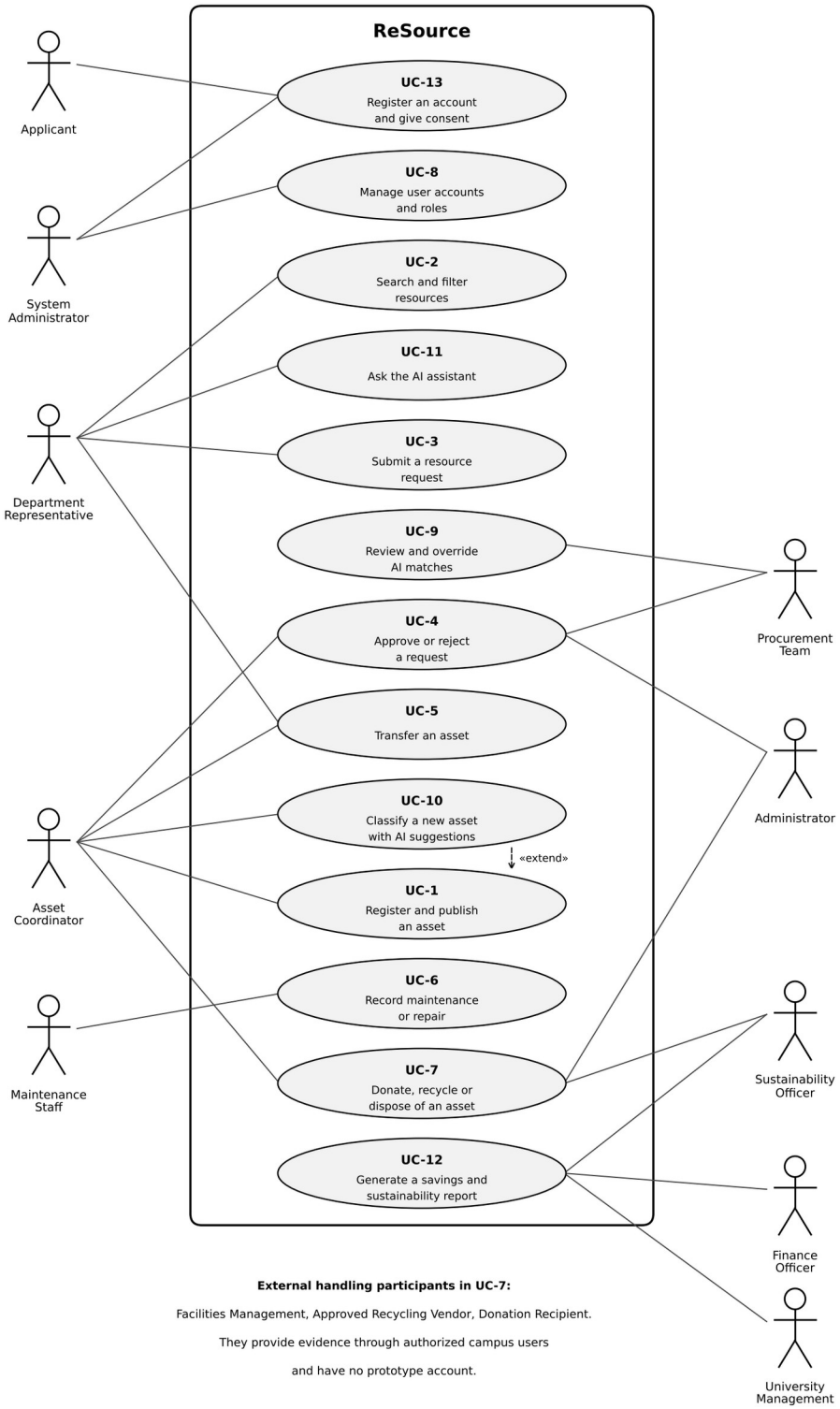


Figure 5.1: Combined ReSource use case diagram. AI components are inside the system boundary; signing in is a precondition, not a separate business use case.

5.3 Use Case Descriptions

The Related requirements row of each use case lists every requirement that names that use case in its Use case column (Sections 2 to 4), so the two directions always agree with each other and with Appendix D. Requirements that apply to every use case are listed once here instead of in each table: FR-6.5, FR-6.7; FR-7.A1, FR-7.A2, FR-7.A5, FR-7.A6; PER-1; SAF-8; SEC-1, SEC-3, SEC-4; QUA-2, QUA-3, QUA-5, QUA-6, QUA-8; BR-12; LEG-4.

UC-2: Search and Filter Resources

Use case ID and name	UC-2: Search and Filter Resources
Description	A Department Representative finds published resources that could meet the department’s need, inspects their current details and optionally continues to a request.
Primary actor (initiator)	Department Representative
Secondary actors	None. The Asset Coordinator maintains the published information but does not act during a search.
Related requirements	FR-1.9; FR-2.1 to FR-2.8; FR-5.9, FR-5.19, FR-5.27; PER-2, PER-6; SAF-1, SAF-5; SEC-5; QUA-7; BR-2.
Preconditions	1. The Department Representative is signed in with an active account and an assigned department. 2. The catalogue is accessible; it may contain no matching assets.
Trigger	The Department Representative opens the catalogue or submits a search.
Main flow	1. The Department Representative enters keywords or an asset ID and selects any filters. 2. The system validates the criteria and applies permissions and all selected filters. 3. The system shows matching published assets with their current status and available quantity. 4. The Department Representative opens a result. 5. The system shows the current permitted asset details and any restrictions. 6. The Department Representative may choose Request, which starts UC-3. Searching never submits a request.
Alternative flows	A1 at step 1: No keywords are entered. The system lists the permitted catalogue using any selected filters. A2 at step 3: No asset matches. The system says so, and the Department Representative changes or clears the criteria. A3 at step 5: The asset is unsuitable. The Department Representative returns to the results.
Exception flows	E1 at step 2: A quantity or range is invalid. The system identifies the problem and keeps the valid criteria. E2 at step 5: The asset was withdrawn, became unsafe or is no longer available. The system shows its current permitted status and blocks a request. E3 at any step: Access expires or is revoked. Protected details are denied. E4 at step 2: The search service fails. The system shows an error and allows a

Use case ID and name	UC-2: Search and Filter Resources
	retry without inventing results. An AI outage does not affect standard search.
Postconditions	Success: Matching permitted resources and their current details are displayed. Failure or exit: No request, reservation, approval or asset change is created.
Rules	R1: Searching never reserves an asset. R2: Only published assets are shown; drafts, withdrawn or restricted records and restricted fields are excluded. R3: UC-3 rechecks availability at submission; a displayed result is not a guarantee.

UC-4: Approve or Reject a Request

Use case ID and name	UC-4: Approve or Reject a Request
Description	The owning department's Asset Coordinator reviews a submitted request and releases or declines it. An Administrator then records the final approval or rejection; approval reserves the requested quantity.
Primary actor (initiator)	Asset Coordinator (of the owning department)
Secondary actors	Administrator (final approval); Procurement Team (second approval when FR-4.3 applies); requesting Department Representative (notified of the decision).
Related requirements	FR-3.8, FR-3.9, FR-3.12; FR-4.1 to FR-4.6, FR-4.14; FR-5.16; FR-7.N1 to FR-7.N6; PER-3, PER-6; SAF-2; SEC-12; QUA-4, QUA-7; BR-2 to BR-5.
Preconditions	1. The Asset Coordinator is signed in, manages the requested asset and did not submit the request. 2. The request is Submitted.
Trigger	The Asset Coordinator opens a Submitted request from their review list.
Main flow	1. The system shows the request, the requesting department, the asset's current available quantity and any restrictions. 2. The Asset Coordinator releases the request. 3. The system records the release and sends the request to the Administrator, and to the Procurement Team when a second approval is required (FR-4.3). 4. The Administrator reviews the released request and approves it. 5. The system rechecks authority, separation of duties, safety restrictions and current availability, records Approved and reserves the requested quantity. 6. The system notifies the requesting Department Representative and the Asset Coordinator; the transfer can then be arranged (UC-5).
Alternative flows	A1 at step 2: The Asset Coordinator declines with a reason. The request becomes Rejected and the Department Representative is notified; nothing is reserved. A2 at step 4: The Administrator, or the Procurement Team member at the second approval, rejects with a reason. The request becomes Rejected and nothing is reserved. A3 at step 4: A second approval is required. The request stays Released until a different Procurement Team member approves; continue at step 5 once all

Use case ID and name	UC-4: Approve or Reject a Request
	approvals exist. A4: No decision is made within 10 working days of submission. The request becomes Expired (FR-3.9).
Exception flows	E1 at step 2 or 4: The reviewer submitted the request or already decided another stage. The system refuses the action and logs it. E2 at step 5: The available quantity is now lower than requested, or the asset was withdrawn or became unsafe. Approval is blocked, current details are shown and nothing is reserved. E3 at step 2 or 4: A decline or rejection reason is missing. The decision cannot be saved. E4 at step 5: Saving or audit recording fails. The request stays Released and nothing is reserved.
Postconditions	Success: All required decisions are recorded, the request is Approved and the quantity is reserved. Ownership and location are unchanged. Rejection or failure: The reason is kept, nothing is reserved and the asset is unchanged.
Rules	R1: Release precedes approval. R2: No one approves their own request or fills two approval stages. R3: Availability is checked before approval; units are never over-allocated. R4: Approval reserves units but does not change ownership or location. R5: AI cannot make an approval decision.

UC-8: Manage User Accounts and Roles

Use case ID and name	UC-8: Manage User Accounts and Roles
Description	The System Administrator reviews account applications and maintains authorized accounts, roles and department membership, with consent and an audit trail.
Primary actor (initiator)	System Administrator
Secondary actors	Applicant or account holder (supplies details and is notified); Department Representative (confirms department membership).
Related requirements	FR-6.3, FR-6.4, FR-6.8 to FR-6.12; FR-7.N1 to FR-7.N5; FR-7.A3; PER-3; SEC-5 to SEC-9; BR-4; LEG-1; ORG-3.
Preconditions	1. The System Administrator is signed in with account-management permission. 2. Role and department records exist or can be created through department management. 3. For activation, the applicant has completed UC-13 with recorded consent, and the requested role has recorded authorization (TBD-1, TBD-3).
Trigger	The System Administrator opens a pending application or selects an existing account or department.
Main flow	1. The System Administrator opens the application or account. 2. The system shows the account status, department, current roles, consent record and open work.

Use case ID and name	UC-8: Manage User Accounts and Roles
	3. The System Administrator checks identity, department and the recorded authorization, then selects the approved role and department or a status change and enters the reason. 4. The system validates required details, duplicates, consent, permission limits and dependencies, and shows the change for confirmation. 5. The System Administrator confirms. The system saves the change and the audit entry, and ends affected sessions where required. 6. The system notifies the account holder and shows the outcome.
Alternative flows	A1 before step 1: The System Administrator sends an invitation. The invitee completes UC-13 with their own details and consent before activation. A2 at step 3: The System Administrator creates, renames or deactivates a department. The system keeps its unique ID and history, and blocks deactivation while dependencies remain. A3 at step 3: The System Administrator suspends or closes the account. Active sessions end and open work is flagged for reassignment. A4: The account holder withdraws consent. Access stops and the System Administrator handles the retained data under the approved policy (SEC-8).
Exception flows	E1 at step 4: Consent, required details or authorization is missing. Activation or the privilege change is blocked. E2 at step 4: A university ID, email or department ID is duplicated. The system identifies the conflict without creating another record. E3 at step 4: The change would raise the System Administrator’s own privileges, or combine roles that are not allowed. The action is refused and logged. E4 at step 4: Deactivation would leave a department’s records unmanaged or remove the last active System Administrator. The system requires reassignment first. E5 at step 5: Saving or audit recording fails. Nothing is committed. E6 at step 6: Notification delivery fails. The saved change stays in force and delivery is retried.
Postconditions	Success: The authorized account or department change is saved, audited and applied to later access checks. Failure or cancellation: No unauthorized activation, privilege change or loss of history occurs.
Rules	R1: Consent precedes retention of personal data and activation. R2: Only the System Administrator applies role changes, with recorded authorization. R3: Technical administration gives no transfer or disposal approval power. R4: Passwords and recovery secrets never appear in records or notices.

UC-13: Register an Account and Give Consent

Use case ID and name	UC-13: Register an Account and Give Consent
Description	An Applicant applies for a ReSource account, reads the privacy notice and terms, and gives explicit consent so that the application can be reviewed and activated (UC-8).
Primary actor	Applicant

Use case ID and name	UC-13: Register an Account and Give Consent
(initiator)	
Secondary actors	System Administrator (reviews and activates the application in UC-8).
Related requirements	FR-6.1 to FR-6.4, FR-6.6; FR-7.N1 to FR-7.N5; PER-3; SEC-2, SEC-6 to SEC-8; LEG-1; ORG-3.
Preconditions	The Applicant has no active account, or holds a valid invitation (FR-6.4). The current privacy notice and terms are published.
Trigger	The Applicant opens the registration page or an invitation link.
Main flow	1. The system shows the privacy notice, the terms, the intended data uses and how to withdraw consent. 2. The Applicant enters name, university ID, university email, department and a password. 3. The Applicant ticks the consent choice, which is initially unticked, and submits. 4. The system checks the required fields, the password rules and duplicates, and records consent with the time and notice version. 5. The system saves the application as Pending approval and notifies the System Administrator. 6. The system tells the Applicant that the account can be used after activation (UC-8).
Alternative flows	A1 at step 3: The Applicant does not consent. Nothing is saved, and the system explains that an account cannot be created without consent. A2 at step 1: The Applicant arrived through an invitation. At step 5 the application goes to the System Administrator for activation as in UC-8.
Exception flows	E1 at step 4: A required field is missing, the password breaks SEC-2 or the email is not a permitted university address. The system names the problem and keeps the other entries except the password. E2 at step 4: The university ID or email already has an account. The system refuses the application and offers sign-in or credential recovery (FR-6.6). E3 at step 5: Saving fails. No application or consent record is kept, and the system reports the failure.
Postconditions	Success: A Pending approval application with a consent record exists; the Applicant cannot use asset functions until activation. Failure or cancellation: No personal data is retained.
Rules	R1: Consent is explicit and recorded before any registration data is kept. R2: Applicants cannot choose privileged roles. R3: Only the personal data listed in SEC-7 is collected.

Appendix A: Glossary

Term	Definition
Administrator / System Administrator	Administrator controls delegated business approvals. System Administrator controls technical access and configuration. These are separate roles.
AED	United Arab Emirates dirham, the currency used for all money values.
AI / semantic matching	Artificial intelligence / finding related resource descriptions by meaning rather than only identical words.
Applicant	A person who has applied for an account that a System Administrator has not yet activated.
Approved vendor list	The Administrator-maintained list of recycling and disposal vendors with an unexpired approval (FR-5.25).
Asset	An individually identified resource or a recorded group of equivalent units managed by the system.
Asset Coordinator	The role responsible for the records, availability and custody of the assets held by their department.
Audit trail	Attributable history of actions and decisions. Corrections are linked records; users cannot overwrite past entries.
Available quantity	Units that remain eligible for requests after reservations and other restrictions are considered.
Avoided purchase / net savings	Estimated new-purchase value avoided through reuse / that estimate adjusted for relevant transfer, repair and other recorded costs.
Category default value	A versioned value per asset category (replacement price, storage and handling cost, weight, additional years of use or emission factor) used when no actual value is recorded.
Compatibility score / confidence	A compatibility score (0 to 100) expresses estimated fit to a request. The confidence level (High, Medium or Low, AI-10) expresses how reliable the result is; neither authorizes an action.
Condition / life-cycle state	Condition describes physical quality, such as Good or Poor. Life-cycle state describes operational status, such as Available, In transfer or Under maintenance.
Consent	An explicit recorded choice to the stated data uses, linked to the notice version and time. It is separate from administrator approval of access.
CSV	Comma-separated values, a plain spreadsheet-compatible file format.
Department Representative	An authorized departmental user who searches for resources, submits and tracks requests for their department and confirms receipt.
End-of-life action	An approved donation, recycling, disassembly, retirement or disposal action, with completion evidence.

Term	Definition
End-of-life review flag	A system flag that asks for an end-of-life decision after an asset is closed as not repairable or reaches the maintenance-cost threshold (FR-5.11, FR-5.12).
Handling party	Facilities Management or an approved vendor assigned to handle hazardous or electronic-waste items and issue a certificate.
Handover / receipt	Handover records departure from the originating department. Receipt confirms arrival and condition and permits custody and location to change.
Hazard flag	A field recorded at registration that marks an asset as hazardous or electronic waste.
Human override	An authorized person’s recorded rejection or correction of an AI suggestion, with a reason where required.
KU	Khalifa University.
Model version / rule version	The identifier of the AI model or service and of the ranking or threshold rules that produced a result, recorded for traceability (AI-5).
OWASP	Open Worldwide Application Security Project, publisher of the Top 10 web security risks [13].
PDF	Portable Document Format, used for uploaded documents and downloaded reports.
PDPL	The UAE Personal Data Protection Law, Federal Decree-Law No. 45 of 2021 [6].
Procurement Team	The role that reviews reuse options for unmatched needs and gives the second approval for high-value requests; purchasing decisions stay under university procedures.
Project description role names	The project description [1] uses Asset Custodian, Requester and Procurement Officer. This document uses the Phase 2 roles [5]: Asset Coordinator, Department Representative (who submits requests for their department) and Procurement Team.
Publication	Making a validated, eligible asset visible in the authorized shared catalogue.
RBAC	Role-based access control: permissions depend on assigned role, department and record scope.
Release / approval	Release is the owning Asset Coordinator’s agreement to make the asset available for the request. Approval is the following authorized business decision, which reserves the units.
Requirement and item prefixes	FR functional requirement; AI general AI requirement; PER performance; SAF safety; SEC security; QUA quality; BR business rule; LEG legal; ORG organizational; UC use case; TBD decision to be determined.
Reservation	Units set aside for an approved request. Submitting a request does not reserve units; the reservation is made at final approval and ends at receipt, cancellation or expiry. It does not change ownership or location.

Term	Definition
ReSource	The proposed campus resource exchange and asset life-cycle management system.
Safety review	A recorded check by a Sustainability Officer that confirms or corrects the hazard classification before a hazardous asset is reused or an end-of-life action proceeds.
SRS	Software Requirements Specification: the documented behavior and constraints used for design and testing.
Waste diversion / CO_2e	Material retained in use or routed away from disposal under the stated method / carbon dioxide equivalent, a unit for estimated climate impact.
WCAG	Web Content Accessibility Guidelines, the W3C accessibility standard used in QUA-2 and QUA-3 [14].
Working day	A day counted by the approved university calendar; the prototype calendar and deadlines require confirmation in TBD-5.

Appendix C: To Be Determined List

Each item has a named team coordinator who will obtain confirmation from the stated stakeholder. Ownership identifies the follow-up responsibility; it does not imply that the stakeholder has been contacted or has approved the proposal. Open items must be resolved or explicitly accepted as prototype assumptions before their stated milestone.

TBD ID	Item to confirm	Where used	Owner	Status
TBD-1	Confirm the named business owner, delegated approvers, role assignments and reuse-first policy through real stakeholder review. Appendix E simulations do not establish approval.	1.3; 3.5; 5.1	Mohamed Almarzooqi with Facilities Management and IT	Open (before requirements sign-off)
TBD-2	Confirm the high-value approval limit, treatment of unknown value, distinct approvers, repair-spending authority and end-of-life delegation. No threshold is claimed as university policy.	FR-4.3; BR-3, BR-7, BR-8	Mohammad Alsuwaidi with Procurement, Finance and Facilities Management	Open (before approval-rule configuration)
TBD-3	Confirm account identity checks, permitted email domains, recovery method, role-authorizing contacts and stricter IT authentication rules. Prototype local accounts remain the baseline; university single sign-on is future work.	FR-6; SEC-2 to SEC-4; ORG-3; UC-13	Mohamed Almarzooqi with IT Department	Open (before access testing)
TBD-4	Approve privacy-notice wording, applicable lawful bases, consent evidence, withdrawal response, retention periods (including audit entries), anonymization and erasure. Reconcile retained audit history with data-removal requests. The 24-hour erasure period in SEC-8 is a proposed prototype default, not a statutory rule.	FR-6.2, FR-6.11; SEC-8; FR-7.A5; 4.1; LEG-1	Rashid Alzaabi with IT and the university privacy/legal contact	Open (before any real personal data)
TBD-5	Confirm working days, holidays, reservation expiry and reminder rules. Ten working days for expiry and five for receipt reminders are proposed prototype defaults, not verified campus policy.	FR-3.9; FR-4.13; FR-7.N6	Mohammad Alsuwaidi with Facilities Management	Open (before deadline tests)
TBD-6	Confirm email delivery service and addresses, failure escalation and channels for inactive users. In-application notifications are essential; email copies are desirable.	FR-7.N; SEC-3	Mohamed Almarzooqi with IT Department	Open (before notification testing)

TBD ID	Item to confirm	Where used	Owner	Status
TBD-7	Confirm maintenance and end-of-life terminology, return-to-service authority, hazardous-item reviewer qualifications, approved handlers and required evidence. Confirm the 50% repair-cost review default (FR-5.11) and the 90-day inspection window for risky transfers (SAF-9).	FR-5; SAF-1 to SAF-9; Appendix B	Rashid Alzaabi with Maintenance Staff and Facilities Management	Open (before workflow sign-off)
TBD-8	Approve financial and environmental formulas, baseline definitions, replacement prices, weights, emission factors and sources. Confirm treatment of repeated reuse, double counting and zero denominators; validate estimates with reviewed examples.	FR-13; Appendix A	Rashid Alzaabi with Finance and Sustainability Officers	Open (before indicator acceptance tests)
TBD-9	Select AI services, model versions, data locations, permitted input fields and fallback settings; obtain IT approval before sending any real university data.	AI-5 to AI-10; SEC-11; LEG-3	Ahmed AlAbdouli with IT Department	Open (before AI integration)
TBD-10	Confirm role-level visibility of department, financial and maintenance data, including who may enter repair costs. Default access is limited by SEC-5, and BR-11 sets who may approve reports with financial figures.	FR-2.5; FR-7.R3; FR-12; FR-13.11	Mohamed Almarzooqi with Finance and IT	Open (before permission-matrix approval)
TBD-11	Confirm university IT governance, real-pilot approval, need and scope of a privacy impact assessment, code and report ownership, licensing and approved terms of use.	LEG-1 to LEG-5; ORG-1, ORG-2	Rashid Alzaabi with university IT and legal contacts	Open (before any real pilot)
TBD-12	Confirm prototype workload, reviewed evaluation datasets, measurable acceptance targets and the AI confidence thresholds in AI-10. The document uses the 60-minute recovery target in QUA-6, and SAF-8 sets the 24-hour backup interval.	PER-1 to PER-6; SAF-8; QUA-1 to QUA-15; AI-10	Ahmed AlAbdouli with all team members and lab instructors	Open (before test-plan baseline)
TBD-13	Confirm category lists, permitted file formats and limits, AI-report change alert (proposed 25%) and upload restrictions. Image/PDF uploads of at most 10 MB remain the draft baseline.	FR-1.6; FR-12.8; SEC-13	Mohammad Alsuwaidi with Rashid Alzaabi	Open (before detailed design)

TBD ID	Item to confirm	Where used	Owner	Status
TBD-14	Confirm the sections required for the Phase 3 deliverable. Resolved: the Phase 3 lab slides [3] require the introduction, the template's system features and nonfunctional requirements (template Sections 4 and 5.1 to 5.4), one use case diagram and the use case descriptions. The template's Sections 2 and 3 are omitted and the remaining sections are numbered in order (1.2).	1.2; 1.5	Mohamed Almarzooqi with lab instructors	Closed
TBD-15	Confirm with real stakeholders the role names, state lists, use cases and requirement priorities agreed by the team. The Appendix E interviews were AI simulations, and real stakeholder interviews remain outstanding.	Sections 2 to 5; Appendices B, D and E	Mohamed Almarzooqi coordinates all four members	Open (before the design baseline)

AI Contributions

Mohamed Almarzooqi

An AI assistant was used to help draft and organize the introduction, FR-2, FR-6, the notification requirements (FR-7.N), the business rules (3.5), the actors list, the use case diagram, UC-2, UC-4, UC-8 and UC-13, the glossary and the TBD list, and to check them against the project description and the Phase 2 report. All content was reviewed and edited by the author.